

Guitar Tap

User Manual

Version 1.0

iOS/iPadOS · macOS · Windows · Linux

Acoustic resonance analysis for luthiers and tonewood researchers.

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See Chapter 6 for a complete reference to spectrum chart gestures and interactions.

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Problem	Likely cause	Fix
No tap detected	Threshold too high	Lower Threshold slider
False detections	Threshold too low or noisy room	Raise Threshold; use Pause between taps
Peaks missing	Peak Min too high	Lower Peak Min slider
Unknown peaks everywhere	Analysis range too wide	Narrow analysis range in Advanced settings
Spectrum doesn't update	Microphone permission not granted	Check System Settings (macOS) or iOS Settings
Plate calculation missing	Dimensions not entered	Open Settings → Measurement Type → Material
Gore thickness not shown	FLC not captured or not enabled	Enable Measure FLC in Settings
Results panel not visible	macOS: window too narrow	Widen the window

Chapter 11 — Glossary

Air (Helmholtz) mode — The resonance of the air cavity inside the instrument body, excited by the motion of air through the sound hole. Also called the Helmholtz resonance.

Bandwidth (–3 dB) — The width of a spectral peak measured at 3 dB below its maximum. Narrower bandwidth (higher Q) indicates lower damping.

Cross/Long ratio — The ratio of cross-grain Young's modulus to longitudinal Young's modulus (E_C / E_L). For spruce, typically 0.04–0.08.

Decay time — The time (in seconds) for a tap response to decay to 1/e of its peak amplitude. A measure of structural damping; higher values indicate lower damping.

Dipole mode — A bending mode of the guitar top plate where the upper and lower bouts move in opposite phase.

E_C (Young's modulus, cross-grain) — Stiffness of the wood measured perpendicular to the grain direction (radial or tangential). Expressed in GPa.

E_L (Young's modulus, longitudinal) — Stiffness of the wood measured along the grain direction. Expressed in GPa. The primary stiffness parameter for tonewood selection.

FFT (Fast Fourier Transform) — The algorithm used to convert the time-domain microphone signal into the frequency-domain spectrum displayed by the app.

FLC tap — A third tap measurement (Flat Long/Cross) used in Plate mode to estimate the shear modulus, which is required for the Gore target thickness calculation.

Free-free beam — A beam supported only at its nodal points (22% from each end), leaving it free to vibrate without damping at the supports. The standard suspension technique for Plate and Brace mode measurements.

Gore target thickness — The recommended final thickness for a guitar top or back plate, calculated from the plate dimensions, Young's moduli, and a stiffness target, following the method described by Trevor Gore.

Q factor — Quality factor; the ratio of a resonant peak's centre frequency to its –3 dB bandwidth. $Q = f_0 / \Delta f$. Higher Q indicates lower damping.

Radiation ratio (R) — A figure of merit for tonewood; $R = c_L / \sqrt{\rho} = \sqrt{(E_L/\rho^3)}$. Higher values indicate a lighter, stiffer wood that radiates sound more efficiently.

Ring mode — A symmetric bending mode of the guitar top plate involving the rim of the lower bout.

Specific modulus (E/ρ) — Young's modulus divided by density. Also written as c^2 (where c is the speed of sound in the material). A key predictor of tap tone quality.

Speed of sound (c) — The speed at which a longitudinal stress wave travels through the wood: $c_L = \sqrt{(E_L / \rho)}$. Expressed in m/s.

Tap Tone Ratio — The ratio of the top mode frequency to the air mode frequency (f_{top} / f_{air}). Typical range for a well-voiced guitar: 1.8–2.4.

Top / Back mode — The fundamental bending mode of the guitar top plate or back plate, respectively; often the strongest peak in a guitar tap spectrum.

Unknown mode — A detected spectral peak that does not fall within any defined frequency window for the selected guitar type. Shown with a "?" label when Show Unknown Modes is enabled.

Upper modes — Resonant modes above the main top and back modes; their significance increases with frequency and build stage.

Appendix A — Keyboard Shortcuts (macOS)

Shortcut	Action
⌘,	Settings
⌘S	Save Measurement…
⌘E	Export Spectrum Image…
⇧⌘E	Export PDF Report…
⌘0	Auto dB
⌘`	Cycle Annotations
⌘M	Show Metrics
⌘L	Show Measurements

> **Note —** Keyboard shortcuts apply to the macOS native app only. The Python desktop app on Windows and Linux uses equivalent menu items; shortcut keys may differ by platform.

Appendix B — File Formats

.guitartap

A JSON-based measurement container. Contains:

- All detected peaks (frequency, magnitude, Q, bandwidth, mode assignment, annotation offset, manual overrides)
- Spectrum snapshot: Base64-encoded little-endian float32 array of magnitude values
- Settings at time of capture (measurement type, frequency windows, Peak Min, analysis range)
- Metadata: capture date, device, location label, notes

Compatible between iOS, iPadOS, and macOS versions of GuitarTap. Can be round-tripped: exported on iOS, imported on macOS, and vice versa.

Calibration Files (.txt / .cal)

Frequency response compensation data for a measurement microphone. Format: two-column plain text, one frequency–magnitude pair per line.

```
<h1 id="frequency-hz-magnitude-db">Frequency (Hz)   Magnitude (dB)</h1>
20.0                -1.2
50.0                -0.4
100.0                0.0
...
20000.0             -3.1
```

The app applies the inverse of the calibration curve to the incoming spectrum, correcting for microphone frequency response deviations.

Exported Formats

Format	Extension	Contents
Measurement data	.guitartap	Full peak data + spectrum (JSON)
Spectrum image	.png	Chart as rendered on screen
PDF report	.pdf	Spectrum chart + peak table + analysis summary